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PiperABM: A Python Library for Resilience-Based Agent Modeling

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Summary

PiperABM is an open-source Python library for building agent-based simulations that couple human behavior with degrading infrastructure on real-world geospatial layouts. In PiperABM, infrastructure includes homes, markets, and the roads connecting them. The library provides lightweight primitives to enable agents with a customizable decision loop, infrastructure with spatial graphs whose effective travel costs evolve, and built-in analysis for accessibility to resources and travel distance, plus optional animation. Because it is implemented with familiar scientific tools like NumPy and NetworkX, researchers can inspect and extend models directly, integrate empirical data, and reuse standard workflows. PiperABM supports resilience research by examining how people's choices and infrastructure conditions co-evolve, seeking to reduce the setup overhead typical of one-off models while remaining flexible enough for diverse hazards, places, and policies.

Statement of Need

One established definition describes resilience as a community's capacity to withstand, adapt to, and recover from disruptions while maintaining or restoring essential functions (United Nations Office for Disaster Risk Reduction, 2017). PiperABM provides a flexible foundation for resilience research: users can adopt study-specific definitions and metrics, customize agent behavior and infrastructure dynamics, and analyze simulation outputs accordingly.

Agent-based models (ABMs) represent systems "from the bottom up" using heterogeneous, interacting agents whose local rules and behaviors generate macro-level patterns (Joshua M. Epstein, 1999; Joshua M. Epstein & Axtell, 1996; Schelling, 2006). ABM has also been highlighted as a framework that engages diverse stakeholders, enables the forecasting of equitable futures, and empowers policy (Hicks et al., 2021). Over recent decades, ABM has matured into a practical research method supported by the scientific Python ecosystem (Hoeven et al., 2025). This paradigm is well-suited to infrastructure and community resilience, particularly within the Food–Energy–Water (FEW) nexus, where access to essential resources depends on both physical networks (e.g., roads linking homes to markets) and human decisions, and where nonlinear feedbacks, thresholds, and cascades are common (Cansino-Loeza et al., 2022; Xue et al., 2024; Zhang et al., 2019).

Resilience studies require graded performance and progressive degradation, such as road deterioration that increases travel costs (Altarabsheh et al., 2021; Llopis-Castelló et al., 2024). They also require two-way feedback between infrastructure and human behavior (Anshuka et al., 2022; Michaelis et al., 2020), and geospatial layouts for place-based analysis (Wang et al., 2022). Reproducible ensemble experiments need consistent reporting and data handling for parameter sweeps, stochastic replications, and uncertainty and sensitivity analysis (Grimm et al., 2020; McCulloch et al., 2022; Razavi et al., 2021). Accessible model state supports

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custom metrics, including spatiotemporal resource accessibility under disruption (Enderami et al., 2024; Tariverdi et al., 2023).

PiperABM is built to address these needs and reduce the setup effort required for place-based resilience studies. PiperABM has been used to investigate how infrastructure degradation, household size, and disruption patterns affect resource accessibility and social-network robustness in an Arctic community context (Noorghasemi, Xiao, & McComb, 2025; Noorghasemi, Xiao, Sha, et al., 2025).

Intended users are researchers and practitioners in infrastructure and community resilience, including urban planners, policymakers, and decision-makers who need transparent, reproducible, place-based simulations.

Comparison to Existing Tools

Among general frameworks, Mesa provides agent definition, scheduling, data collection, and browser visualization (Hoeven et al., 2025); modelers supply infrastructure degradation, accessibility metrics, and behavior–infrastructure coupling. InfraRisk models hazards, cascades, and recovery in interdependent power, water, and transport networks (Balakrishnan & Cassottana, 2022), with human decision-making typically added separately. Repast4Py supports distributed and parallel ABMs (Collier & Ozik, 2022), leaving resilience-specific logic to modelers. PiperABM integrates continuous degradation, resource accessibility, and coupled behavior–infrastructure dynamics with customizable decision rules, reducing setup effort for resilience studies.

Purpose-built models address multilayer recovery after Hurricane Harvey (Xue et al., 2024), household recovery after Sandy (RecovUS) (Moradi & Nejat, 2020), storm-induced power-outage restoration with crew dispatch (Walsh et al., 2018), community resilience under tornado hazards (Aghababaei & Koliou, 2023), and flood risk–insurance dynamics (Dubbelboer et al., 2017). These models serve specific studies, and adapting them to other settings often requires reimplementing similar components. PiperABM provides reusable components and extension points for adapting models to new hazards, geographies, and policies, while supporting consistent scenario comparisons and analysis with standard Python tools.

Model Overview

PiperABM couples infrastructure and society networks represented as NetworkX graphs, with their interactions summarized in Figure 1.

The infrastructure network represents the physical layout: nodes represent homes, markets, and junctions; edges represent streets and access connections from homes and markets to streets. Spatial layouts can be constructed manually or imported from map data.

Society nodes represent individual agents, with family, neighbor, and friendship ties as edges. Family ties connect agents assigned to the same home. Agents have a home assignment, current location, monetary balance, socioeconomic status, resource inventories, and sufficiency thresholds, with food, water, and energy as the default resources. Home assignments and agent locations link the two networks. Markets hold resource inventories, while social ties identify friends' and neighbors' homes as potential destinations.

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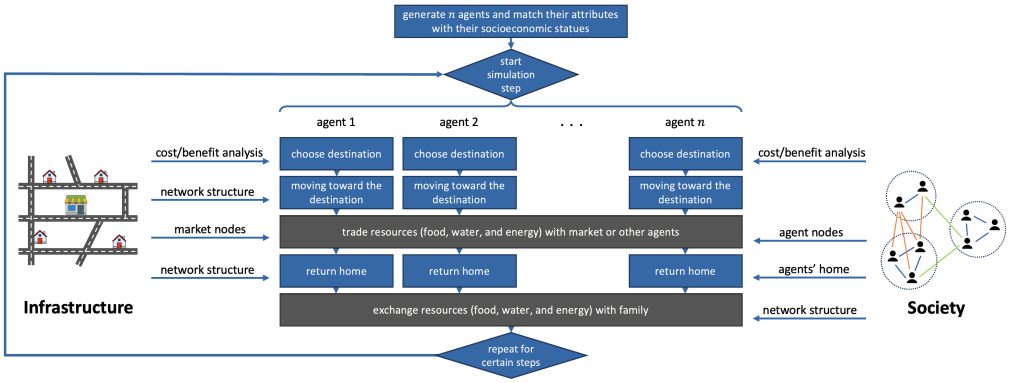


Figure 1: PiperABM overview: Infrastructure locations and social relationships shape agents' travel and resource exchanges.

The baseline decision mechanism follows OODA-style logic: observing, orienting, deciding, and acting (Brehmer, 2005; Johnson, 2023). When an agent's queued actions finish, it selects a new destination, considering markets first and socially connected homes if no suitable market is found. Destinations are ranked by resource value minus estimated travel resource costs, with travel-time constraints determining feasibility. Agents queue travel, a stay at the destination, a return home, and rest. At each simulation step, the model coordinates resource exchanges among agents at the same home or with markets where agents are present, using a cooperative, centralized Nash bargaining formulation (Nash, 1950). Market inventories are replenished each step.

Agents consume resources while staying and traveling, and travel accumulates usage on infrastructure edges. With degradation enabled, usage and aging increase streets' effective lengths, which determine A* routing costs (Foead et al., 2021) and travel duration. Infrastructure conditions therefore affect subsequent routes and feasible destinations, while agent movement contributes to further degradation.

Outputs include travel distance and resource accessibility, measured as inventory relative to a sufficiency threshold and capped at one. KeepDelta can store human-readable changes in model state (Noorghasemi & McComb, 2025), and optional animation supports face-validity assessment. Researchers can replace the baseline decision and degradation rules through user-defined extensions.

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References

Aghababaei, M., & Koliou, M. (2023). Community resilience assessment via agent-based modeling approach. *Computer-Aided Civil and Infrastructure Engineering*, *38*(7), 920–939. https://doi.org/10.1111/mice.12916

Altarabsheh, A., Altarabsheh, R., Altarabsheh, S., & Asi, I. (2021). Prediction of pavement performance using multistate survival models. *Journal of Transportation Engineering, Part B: Pavements*, *147*(1), 04020082. https://doi.org/10.1061/JPEODX.0000241

DRAFT

Anshuka, A., Ogtrop, F. F. van, Sanderson, D., & Leao, S. Z. (2022). A systematic review of agent-based model for flood risk management and assessment using the ODD protocol. *Natural Hazards*, *112*(3), 2739–2771. https://doi.org/10.1007/s11069-022-05286-y

Balakrishnan, S., & Cassottana, B. (2022). InfraRisk: An open-source simulation platform for resilience analysis in interconnected power–water–transport networks. *Sustainable Cities and Society*, *83*, 103963. https://doi.org/10.1016/j.scs.2022.103963

Brehmer, B. (2005). The dynamic OODA loop: Amalgamating boyd's OODA loop and the cybernetic approach to command and control. *Proceedings of the 10th International Command and Control Research Technology Symposium*, 365–368.

Cansino-Loeza, B., Munguía-López, A. del C., & Ponce-Ortega, J. M. (2022). A water-energy-food security nexus framework based on optimal resource allocation. *Environmental Science & Policy*, *133*, 1–16. https://doi.org/10.1016/j.envsci.2022.03.006

Collier, N., & Ozik, J. (2022). Distributed agent-based simulation with Repast4Py. *2022 Winter Simulation Conference (WSC)*, 192–206. https://doi.org/10.1109/WSC57314.2022.10015389

Dubbelboer, J., Nikolic, I., Jenkins, K., & Hall, J. (2017). An agent-based model of flood risk and insurance. *Journal of Artificial Societies and Social Simulation*, *20*(1), 6. https://doi.org/10.18564/jasss.3135

Enderami, S. A., Sutley, E., Helgeson, J., Dueñas-Osorio, L., Watson, M., & Lindt, J. W. van de. (2024). Measuring post-disaster accessibility to essential goods and services: Proximity, availability, adequacy, and acceptability dimensions. *Journal of Infrastructure Preservation and Resilience*, *5*(1). https://doi.org/10.1186/s43065-024-00104-0

Epstein, Joshua M. (1999). Agent-based computational models and generative social science. *Complexity*, *4*(5), 41–60. https://doi.org/10.1002/(sici)1099-0526(199905/06)4:5%3C41::aid-cplx9%3E3.0.co;2-f

Epstein, Joshua M., & Axtell, R. (1996). *Growing artificial societies: Social science from the bottom up*. Brookings Institution Press; MIT Press. ISBN: 9780262550253

Foead, D., Ghifari, A., Kusuma, M. B., Hanafiah, N., & Gunawan, E. (2021). A Systematic Literature Review of A* Pathfinding. *Procedia Computer Science*, *179*, 507–514. https://doi.org/10.1016/j.procs.2021.01.034

Grimm, V., Railsback, S. F., Vincenot, C. E., Berger, U., Gallagher, C., DeAngelis, D. L., Edmonds, B., Ge, J., Giske, J., Groeneveld, J., Johnston, A. S. A., Milles, A., Nabe-Nielsen, J., Polhill, J. G., Radchuk, V., Rohwäder, M.-S., Stillman, R. A., Thiele, J. C., & Ayllón, D. (2020). The ODD protocol for describing agent-based and other simulation models: A second update to improve clarity, replication, and structural realism. *Journal of Artificial Societies and Social Simulation*, *23*(2), 7. https://doi.org/10.18564/jasss.4259

Hicks, S., McComb, C., & Alessa, L. (2021). *Agents and the Arctic: The Case for Increased Use of Agent-Based Modeling to Study Permafrost — thearcticinstitute.org*. https://www.thearcticinstitute.org/agents-arctic-case-increased-use-agent-based-modeling-study-permafrost/.

Hoeven, E. ter, Kwakkel, J., Hess, V., Pike, T., Wang, B., rht, & Kazil, J. (2025). Mesa 3: Agent-based modeling with Python in 2025. *Journal of Open Source Software*, *10*(107), 7668. https://doi.org/10.21105/joss.07668

Johnson, J. (2023). Automating the OODA loop in the age of intelligent machines: Reaffirming the role of humans in command-and-control decision-making in the digital age. *Defence Studies*, *23*(1), 43–67. https://doi.org/10.1080/14702436.2022.2102486

Llopis-Castelló, D., Camacho-Torregrosa, F. J., Romeral-Pérez, F., & Tomás-Martínez, P. (2024). Estimation of pavement condition based on data from connected and autonomous

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vehicles. *Infrastructures*, *9*(10). https://doi.org/10.3390/infrastructures9100188

McCulloch, J., Ge, J., Ward, J., Heppenstall, A., Polhill, J. G., & Malleson, N. (2022). Calibrating agent-based models using uncertainty quantification methods. *Journal of Artificial Societies and Social Simulation*, *25*(2), 1. https://doi.org/10.18564/jasss.4791

Michaelis, T., Brandimarte, L., & Mazzoleni, M. (2020). Capturing flood-risk dynamics with a coupled agent-based and hydraulic modelling framework. *Hydrological Sciences Journal*, *65*(9), 1458–1473. https://doi.org/10.1080/02626667.2020.1750617

Moradi, S., & Nejat, A. (2020). RecovUS: An agent-based model of post-disaster household recovery. *Journal of Artificial Societies and Social Simulation*, *23*(4), 13. https://doi.org/10.18564/jasss.4445

Nash, J. F. (1950). The Bargaining Problem. *Econometrica*, *18*(2), 155–162. https://doi.org/10.2307/1907266

Noorghasemi, A., & McComb, C. (2025). KeepDelta: A Python Library for Human-Readable Data Differencing. *Journal of Open Source Software*, *10*(110), 8075. https://doi.org/10.21105/joss.08075

Noorghasemi, A., Xiao, M., & McComb, C. (2025). *Evaluating community well-being and resilience via agent-based models*. https://doi.org/10.22541/essoar.173687385.52831340/v1

Noorghasemi, A., Xiao, Y., Sha, Z., Xiao, M., & McComb, C. (2025). *Agent-based modeling for the evaluation of community resilience in silico*. https://doi.org/10.22541/au.176229737.71866255/v1

Razavi, S., Jakeman, A., Saltelli, A., Prieur, C., Iooss, B., Borgonovo, E., Plischke, E., Lo Piano, S., Iwanaga, T., Becker, W., Tarantola, S., Guillaume, J. H. A., Jakeman, J., Gupta, H., Melillo, N., Rabitti, G., Chabridon, V., Duan, Q., Sun, X., … Maier, H. R. (2021). The future of sensitivity analysis: An essential discipline for systems modeling and policy support. *Environmental Modelling & Software*, *137*, 104954. https://doi.org/10.1016/j.envsoft.2020.104954

Schelling, T. C. (2006). *Micromotives and macrobehavior* (Revised). W. W. Norton & Company. https://doi.org/10.2307/134812

Tariverdi, M., Nunez-del-Prado, M., Leonova, N., & Rentschler, J. (2023). Measuring accessibility to public services and infrastructure criticality for disasters risk management. *Scientific Reports*, *13*(1). https://doi.org/10.1038/s41598-023-28460-z

United Nations Office for Disaster Risk Reduction. (2017). *The Sendai Framework Terminology on Disaster Risk Reduction: Resilience*. https://www.undrr.org/terminology/resilience

Walsh, T., Layton, T., Wanik, D., & Mellor, J. (2018). Agent based model to estimate time to restoration of storm-induced power outages. *Infrastructures*, *3*(3). https://doi.org/10.3390/infrastructures3030033

Wang, B., Hess, V., & Crooks, A. (2022). Mesa-Geo: A GIS Extension for the Mesa Agent-Based Modeling Framework in Python. *Proceedings of the 5th ACM SIGSPATIAL International Workshop on GeoSpatial Simulation*, 1–10. https://doi.org/10.1145/3557989.3566157

Xue, J., Park, S., Mondal, W. U., Reia, S. M., Yao, T., & Ukkusuri, S. V. (2024). An agent-based model of post-disaster recovery in multilayer socio-physical networks. *Sustainable Cities and Society*, *115*, 105863. https://doi.org/10.1016/j.scs.2024.105863

Zhang, P., Zhang, L., Chang, Y., Xu, M., Hao, Y., Liang, S., Liu, G., Yang, Z., & Wang, C. (2019). Food-energy-water (FEW) nexus for urban sustainability: A comprehensive review. *Resources, Conservation and Recycling*, *142*, 215–224. https://doi.org/10.1016/j.

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